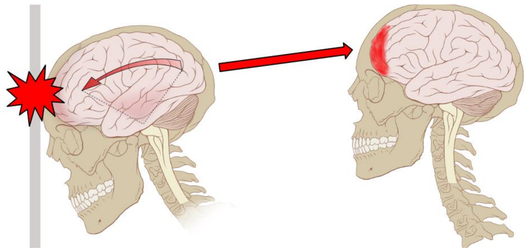
Functional Network Disruption as Biomarkers for Adolescent Concussion

Dawei Wang[a,m]*, Adrienne Davis[e], Avyarthana Dey[a,n], Pavreet Gill[c], Andrea Hickling[c,d], George M Ibrahim[a,f], Alexa Irvin[c], Phillip Johnston[a], Nikou Kelardashti[a], Shannon E Scratch[c,h,i], Jaehyun Sur[b], Anne L Wheeler[a,g], Rouzbeh Zamyadi[a], Benjamin T Dunkley[a,b,j,k,l]*

a. Neurosciences & Mental Health, Hospital for Sick Children Research Institute, Toronto, ON, Canada, b. Pharmacology & Toxicology, University of Toronto, Toronto, ON, Canada, c. Bloorview Research Institute, Holland Bloorview Kids Rehabilitation Hospital, Toronto, ON, Canada, d. Department of Occupational Science and Occupational Therapy, Temerty Faculty of Medicine, University of Toronto, Toronto, ON, Canada, e. Paediatric Emergency Medicine, Hospital for Sick Children, Toronto, ON, Canada, f. Neurosurgery, Hospital for Sick Children, Toronto, ON, Canada, g. Department of Physiology, University of Toronto, Toronto, ON, Canada, h. Department of Paediatrics, University of Toronto, Toronto, ON, Canada, i. Rehabilitation Sciences Institute, University of Toronto, Toronto, ON, Canada, j. Diagnostic & Interventional Radiology, Hospital for Sick Children, Toronto, ON, Canada, k. Medical Imaging, University of Toronto, Toronto, ON, Canada, l. School of Psychology, University of Nottingham, Nottingham, United Kingdom, m. Human Biology, University of Toronto, Toronto, ON, Canada, n. Neurosciences Research Program, St. Michael's Hospital

Introduction



- Concussion and trauma-related injuries affect around 400,000 Canadians annually
- Improper or late diagnosis can lead to long-term changes to cognitive and behavioural health
- Most existing diagnostic approaches over rely on physical symptoms and lack sensitivity and reproducibility
- Concussion is known to heavily disrupt the alpha frequency, the strongest signal in the brain

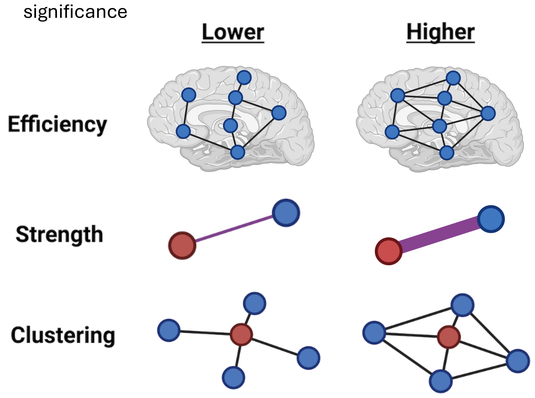
Aims and Hypothesis

- Aims: Model adolescent concussion patients' brain functional connectivity network using **graph theory** to search for changes and possible concussion biomarkers
- **Hypothesis: Brain functional network organization is disrupted in concussion compared to typical controls**

Methods

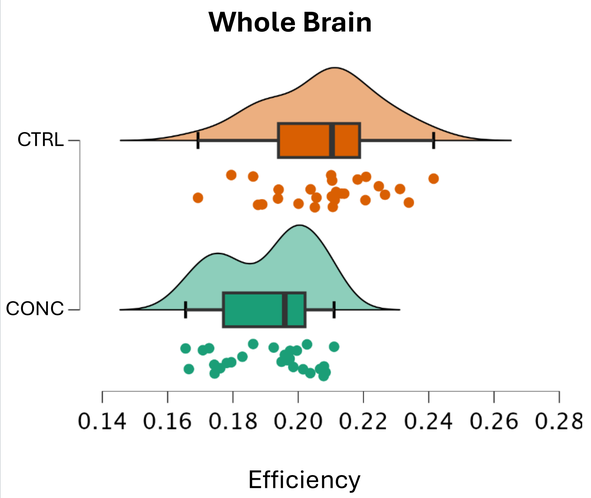
Participant characteristics	Concussion	Typical controls	Statistical tests
Sample size	27	28	-
Age (years)	14.96 ± 1.53	14.58 ± 1.30	N.S.
Sex (M/F)	8/19	9/19	N.S.
Mean time since last concussion (days)	709.3	-	-
Average number of concussions	2	-	-

- Magnetoencephalography scan in 54 regions of interest defined by the **Automated Anatomical Labelling atlas**
- Network analysis performed using **Fieldtrip** and **Brain Connectivity Toolboxes**
- Appropriate tests were used to assess group differences; **Benjamini-Hochberg** procedure used to assess significance



Results

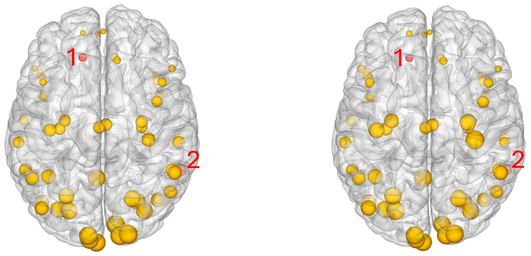
Efficiency



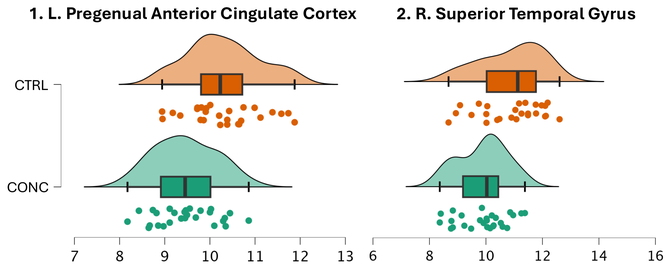
Efficiency Independent Samples T-Test

Test	Statistic	d.f.	p-value
Student	-3.946	53.00	< 0.001
Welch	-3.959	52.03	< 0.001

Strength



Each dot represents the averaged strength value of a brain region; red dot represents most significantly affected node



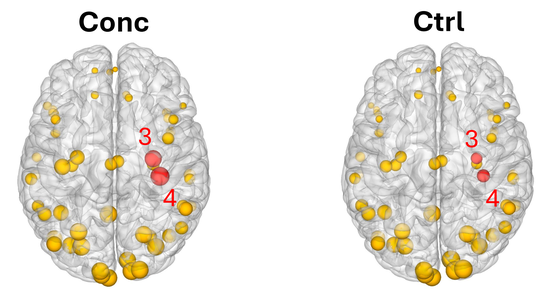
Strengths Independent Samples T-Test (Region 1)

Test	Statistic	d.f.	p-value
Student	-4.160	52.00	<0.001
Welch	-4.160	50.77	<0.001

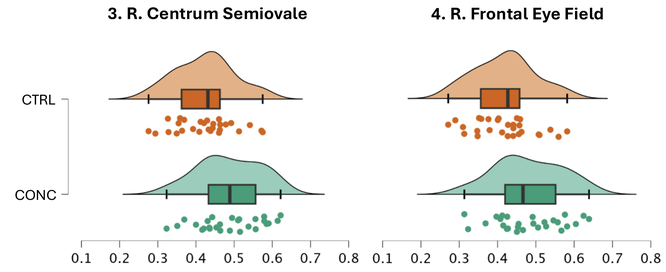
Strengths Independent Samples T-Test (Region 2)

Test	Statistic	d.f.	p-value
Student	-3.706	51.00	<0.001
Welch	-3.688	46.94	<0.001

Clustering



Each dot represents the averaged strength value of a brain region; red dot represents most significantly affected node



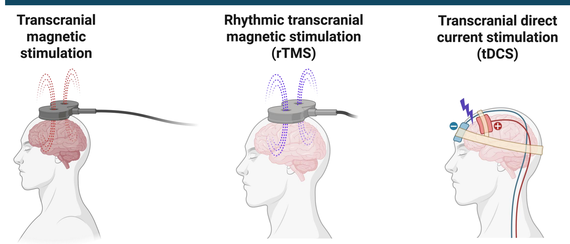
Clustering Independent Samples T-Test (Region 3)

Test	Statistic	d.f.	p-value
Student	2.949	53.00	0.005
Welch	-2.946	52.51	0.005

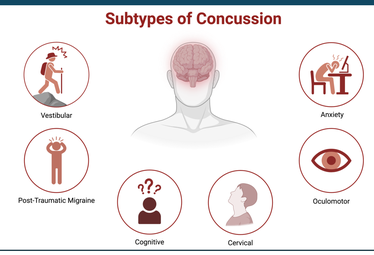
Clustering Independent Samples T-Test (Region 4)

Test	Statistic	d.f.	p-value
Student	2.753	53.00	0.008
Welch	2.748	51.97	0.008

Discussion



- Altered network organization in concussion patients compared to controls
- Disrupted connectivity in alpha frequency is a **potential concussion biomarker**
- **Precision medicine:** affected regions could act as potential targets for neurostimulation therapies
- Mapping allows more specific diagnosis; distinguishing different subtypes
- Earlier and precise diagnoses allows for **more immediate treatment**; preventing worsening symptoms and further brain health deterioration, particularly in **vulnerable demographic such as adolescents and children**



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